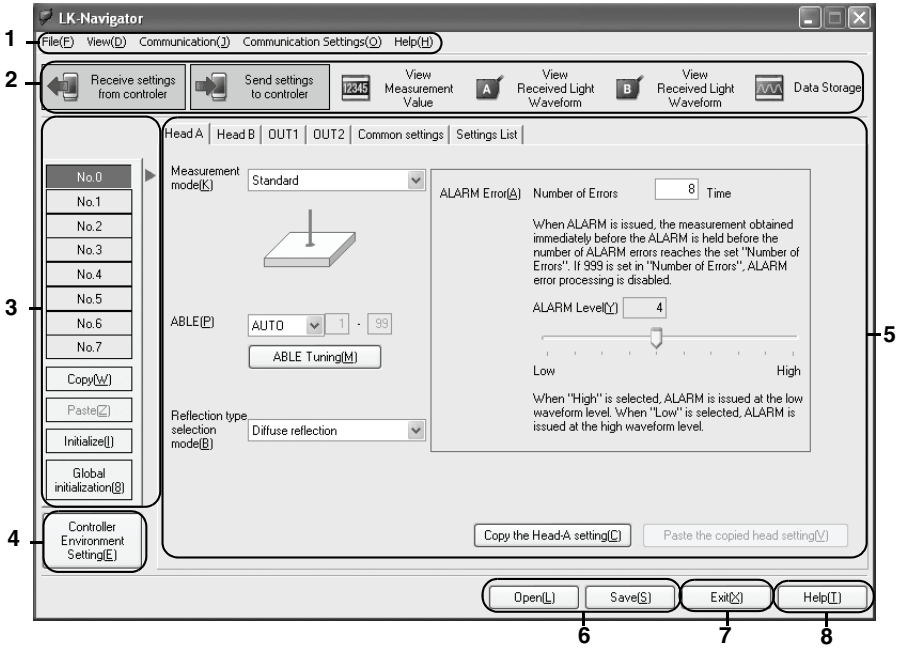
Identifying Part Names and Functions of the Main Screen

The part names and functions on the main screen of "LK-Navigator" are descrived as follows.



1. Menu bar

Shows the operation menu of "LK-Navigator".

2. Tool bar

Shows the frequently used tools of the menu bar in the form of buttons.

For the operating method of the respective tools, refer to "View Measurement Value" (page 3-29) to "Data Storage Setting" (page 3-34).

3. Program Settings

Used to select the desired Program No., to copy it and to initialize it.

For the method of setting the programs, refer to "Setting the Programs" (page 3-26).

4. Controller Envriroment Setting

Sets the operating environment settings of the controller.

For the method of setting the environment settings, refer to "Environment Setting" (page 3-22).

5. Setting Window

Sets the respective heads and their outputs.

For the method of setting the respective tabs, refer to "Basic Settings" (page 3-6).

6. Open or Saving the Setting File

Used to read the settings that are made by "LK-Navigator", and save them.

For the method of reading and saving the setting file, refer to "Reading and Saving the Setting File" (page 3-42).

7. Quit

Quits the "LK-Navigator".

For the method of quitting the LK-Navigator, refer to "Start and Quit" (page 2-11).

8. Help

Opens the Help file of the "LK-Navigator".

Flow of Operations

The procedure to set the Head-A signal to be output from OUT1 when the "LK-Navigator" is started up for the first time after purchase, is described below.

Connect the controller to a PC using the RS-232C cable or USB cable.

Refer to "Connecting the Controller to a PC" (page 2-2).

Start up the controller first, then start up the PC.

Start up LK-Navigator.

Refer to "Start and Quit" (page 2-11).

Select [Start up with the default values] from the [Select the settings to be displayed at startup] window.

Refer to "Start and Quit" (page 2-11).

Select the Program No.

Refer to "Setting the Programs" (page 3-26).

Set the [Common Settings].

Refer to "Common Settings" (page 3-17).

Set [Head-A].

Refer to "Head-A/Head-B" (page 3-6).

Set [OUT 1].

Refer to "OUT1/OUT2" (page 3-10).

Confirm contents of the settings on [Settings List].

Refer to "Settings List" (page 3-20).

Set the method to communicate with the controller using [Controller Environment Setting].

Refer to "Environment Setting" (page 3-22).

Send the settings to the controller.

Refer to "Sending/Receiving the Settings" (page 3-21).

Confirm the operations using the tools.

Refer to "View Measurement Value" (page 3-29) to "Data Storage Setting" (page 3-34).

Save the setting file in a PC.

Refer to "Reading and Saving the Setting File" (page 3-42).

End

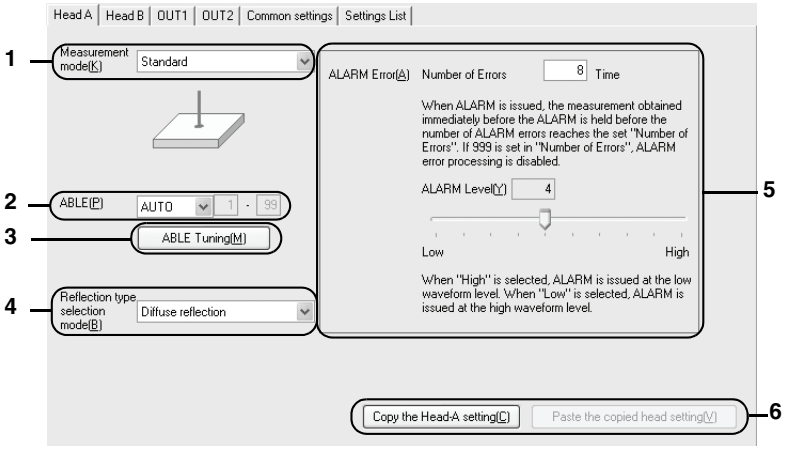
Basic Settings

Basic settings of "LK-Navigator" are described below.

Head-A/Head-B

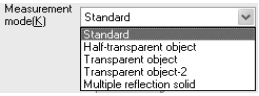
The screen and functions of the Head-A tab and those of the Head-B tab are described below.
The heads that are connected to the controller can be set using this function. The items required to be set are identical in Head-A and Head-B. The settings should be made in accordance with the environment in use.

Click the [Head A] tab or the [Head B] tab on the setting window.



1. Measurement mode

Select the measurement mode.



2. ABLE

Select either automatic (AUTO) or the manual (MANUAL) control of the light intensity.

Reference

- Select [AUTO] for normal operation.
- When [MANUAL] is selected, the upper limit and the lower limit of the light intensity control are set.
  Setting range: 1 to 99

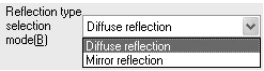
3. ABLE Tuning

Calibrate the settable range of the [MANUAL] mode using the [ABLE] function for optimum performance in accordance with the actual work-piece.

Refer to "Setting Procedure of ABLE Calibration" (page 3-8).

4. Relfection type selection mode

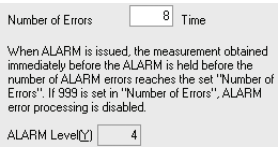
Select the reflection type.



5. Alarm Error

Number of Errors

Set the number of times of the normal measurement obtained before the error is held while an error is occuring.



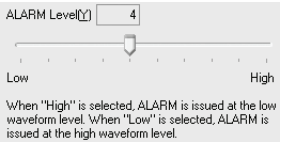
Reference

Setting range: 0 to 999

Number of errors	Error processing
0	Immediate processing of the ALARM error
1 to 998	The normal value immediately before occurrence of the error is HELD.
999	ALARM error processing is not performed.

ALARM Level

Set the alarm threshold level.



Setting range: Low (0) to High (9)

- Low (0): When "Low" is selected, ALARM is issued at the high waveform level, i.e., is not issued easily.
- High (9): When "High" is selected, ALARM is issued at the low waveform level, i.e., is issued easily.

6. Copy the Head-A setting. Paste the copied head setting.

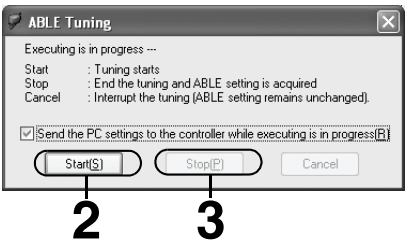
This function is used to copy the already established setting of a specific head, to another head (including the head setting of other program No.)

Refer to "Procedure to Copy the Setting of a Head to Another Head" (page 3-9).

■ Setting Procedure of ABLE Calibration

1 **Click the [ABLE Tuning] button.**

The [Head Setting] window appears.



▶ Note

- If you do not want to send the head setting to the controller at the time when calibration starts, turn OFF the check box [Send the PC setting to the controller while executing is in progress].
- While executing the [ABLE Tuning], the controller enters the "Communication mode".

2 **Click the [Start] button to start calibration.**

Then both the [Stop] button and the [Cancel] button become active.

3 **Click the [Stop] button. The calibration ends and the optimal values are set automatically.**

► Note

If the [Cancel] button is clicked, the calibration is cancelled so that the settings are not changed.

■ Procedure to Copy the Setting of a Head to Another Head

The procedure of copying the [Head-A] setting to [Head-B] is described below as an example.

First, complete all settings of [Head-A].

1 **Click the [Copy the Head-A setting] button.**

The [Paste the copied head setting] button becomes active.

2 **Click the [Head-B] tab.**

Reference

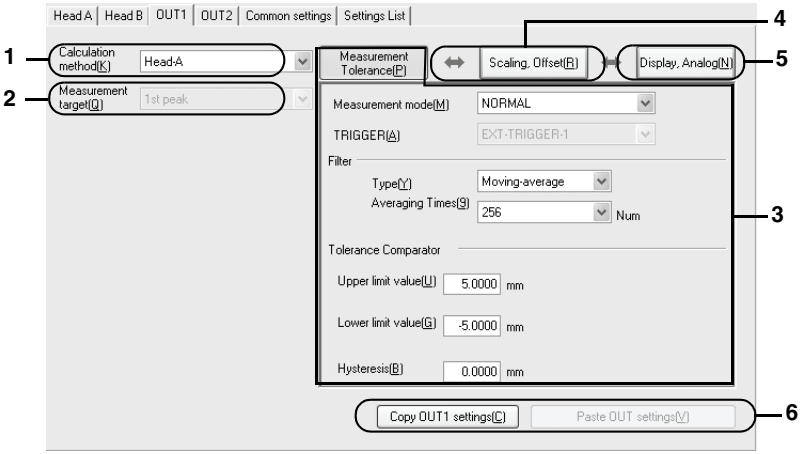
You can paste the setting to the head setting of other Program No. To paste the setting to the head setting of other Program No., select first the desired Progam No., then select the head.

3 **Click the [Paste the copied head setting] button.**

OUT1/OUT2

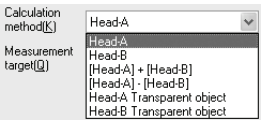
Functions and screens of the OUT1/OUT2 tab are described below.
Use this function to set the processing method that is best suited for the specific measured value. The items required to be set are identical in OUT1 and OUT2.

Click the [OUT1] tab or the [OUT2] tab of the setting window.



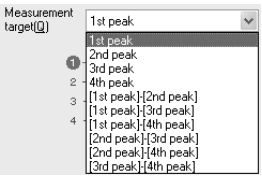
1. Calculation method

Set the calculation method that is best suited for the selected method of measurement and for the specific measurement target.



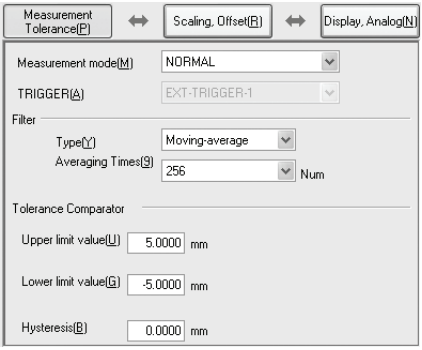
2. Measurement target

When [Head-A transparent object] or [Head-B transparent object] is selected by [Calculation method], the [Measurement target] is required to be set.



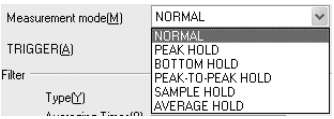
3. Measurement, Tolerance

The method to set the Measurement and to set the Tolerance Comparator is described below.



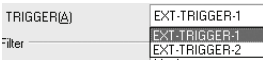
Measurement mode

Set the measurement mode.



TRIGGER

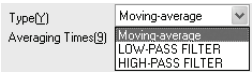
When any item other than [NORMAL] is selected by the [Measurement mode], trigger is requested to be set.



Filter

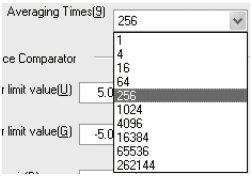
- Type

Use this function to set the filter processing that is best suited for the measured value.



- Averaging Times

Set the number of the Averaging Times when [Moving-average] is selected by [Type].

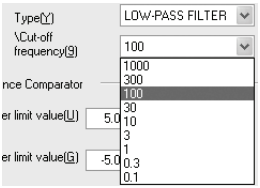


Reference

As the number of Averaging Times decreases, response speed of measured value increases. As the number of Averaging Times increases, a more stable measurement value can be obtained.

- Cut-off frequency

When either [LOW-PASS FILTER] or [HIGH-PASS FILTER] is selected by [Type], its cut-off frequency is requested to be set.



Tolerance Comparator

- Upper limit value

Sets the upper limit value of the Tolerance.

- Lower limit value

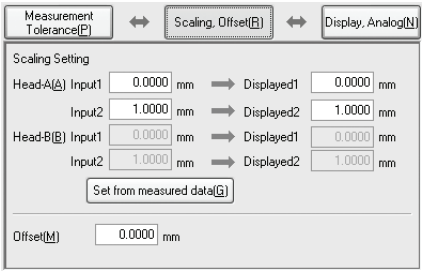
Sets the lower limit value of the Tolerance.

- Hysteresis

Sets hysteresis of the Tolerance Comparator.

4. Scaling, Offset

Set the Scaling and Offset.



Scaling Setting

- Head-A input value-1/Display value-1, Head-A input value-2/Display value-2

The method to display the Head-A input value on the display can be set using the Scaling function at the two measurement points as follows.

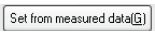
- Head-B input value-1/Display value-1, Head-B input value-2/Display value-2

The method to display the Head-B input value on the display can be set using the Scaling function at the two measurement points as follows.

- Setting from the Measurement Data

Set the Scaling while confirming the actual measurement data.

Refer to "Procedure to Implement Setting from the Measurement Data" (page 3-15).

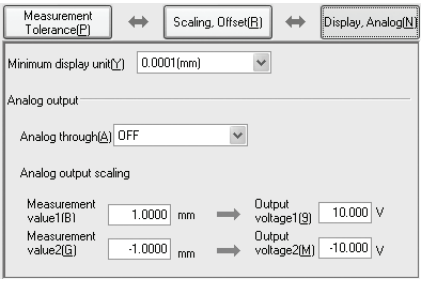


Offset

Set the offset.

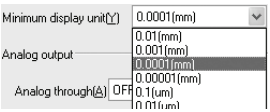
5. Display, Analog

Set the measurement value display and the analog output.



Minimum display unit

Set the decimal point position of the measurement and setting values, and set the number of digits to display.



Reference

The digits that are not displayed below decimal point will be rounded off.

▶ Note

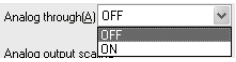
When the setting [Minimum display unit] is changed, settings of the following items are initialized.

Scaling/Offset/Tolerance Setting/Analog Output Scaling/Auto Zero Reference Value

Analog output

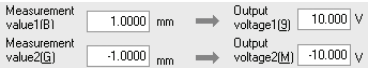
- Analog through

Set whether the measurement value is output to the Analog output with or wihtout processing, regardless of the setting of [Measurement Mode].



- Analog output scaling

Set the Scaling on the Analog output with respect to the measurement value.



Reference

Scaling can be applied to the Analog output in the range of ±10.5 V.

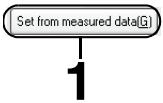
6. Copy OUT-1 settings. Past OUT settings.

This function is used to copy the already established OUT-1's setting to the other OUT's settings (including the OUT setting of other program No.)

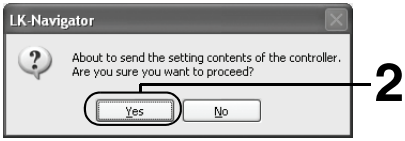
Refer to "The procedure to copy the OUT's setting to the other OUT's setting" (page 3-16).

■ Procedure to Implement Setting from the Measurement Data

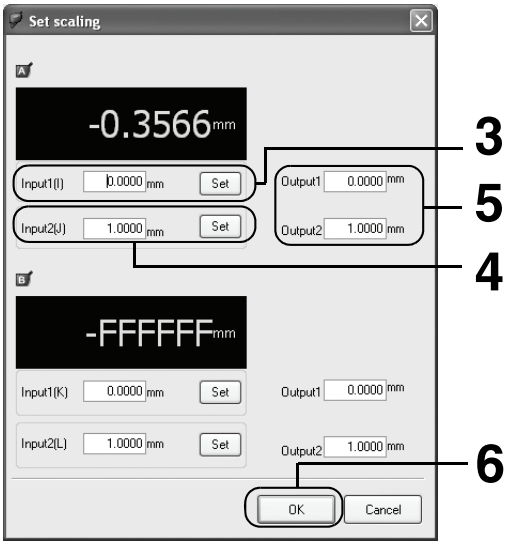
1 **Click the [Set from measured data] button.**



The window prompting you to confirm the send transmission appears.



2 **Click the [Yes] button.**
The [Set scaling] window appears.



▶ Note

While executing [Set scaling], the controller enters the "Communication mode".

3 **Click the [Set] button of [Input 1] when measuring the reference surface.**
The measured value at the reference surface is entered in [Input 1].

4 **Place the Master Work-piece below sensor for measurement, and click the [Set] button of [Input 2].**

The measured value obtained from the Master Work-piece is entered in [Input 2].

5 **Set the Scaling on [Output 1] and [Output 2].**

6 **Click the [OK] button.**

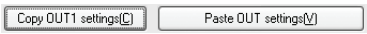
■ The procedure to copy the OUT's setting to the other OUT's setting

Copying the OUT1 setting to OUT2 is described as an example.

Complete all settings of [OUT1].

1 **Click the [Copy the OUT1 settings] button.**

The [Paste OUT settings] button becomes active.



2 **Select the [OUT2] tab.**

Reference

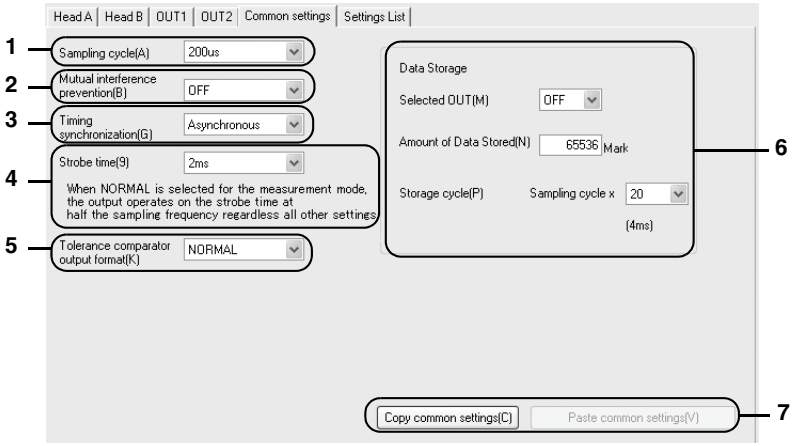
You can copy the OUT's setting to the OUT of any other program No. To paste the OUT's setting to that of other Program No., select first the desired Program No., then select the desired OUT.

3 **Click the [Paste OUT settings] button.**

Common Settings

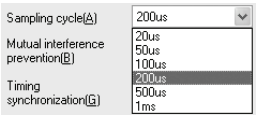
Functions and screen of the Common Settings tab are described below.

Click the [Common Settings] tab of the Setting window.



1. Sampling cycle

Set the sampling cycle of measurements.



2. Mutual interference prevention

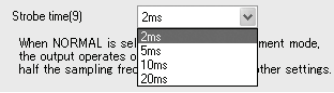
This function is used to prevent mutual interference between two heads when the two heads are connected to the same controller.

3. Timing synchronization

This function is used whether to synchronize the timin inputs of OUT1 and OUT2, or not to synchronize them.

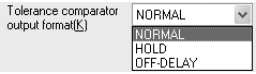
4. Strobe time

When any measurement mode other than NORMAL is selected, the strobe time can be set.



5. Tolereance comparator output format

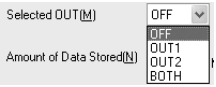
This function is used to set the operation of the Tolerance Comparator Output when result of the Tolerance Comparator judgment changes from ON to OFF.



6. Data Storage

Selected OUT

Set the OUT that becomes the target of data storage.

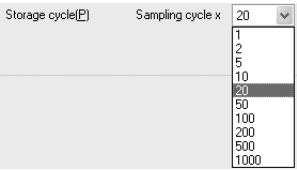


Amount of Data Stored

Set the number of data to be stored.

Storage cycle

Set the interval at which the data is stored.



Reference

Data is stored at the interval (in units of milli second or micro second) of the storage cycle that is displayed under the set value.

7. Copy common settings, Past common settings

This function is used to copy the already established setting of a program to the Common Settings of the other program No.

Refer to "Procedure of Copying the Settings of the Common Settings to Other Programs" (page 3-19).

■ Procedure of Copying the Settings of the Common Settings to Other Programs

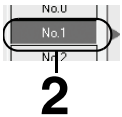
Copying the setting of program No. 0 to [Common Settings] of program No. 1 is described as an example.

Complete all settings of [Common Settings].

1 Click the [Copy common settings] button of program No. 0.

The [Paste common settings] button becomes active.

2 Select the program No.1.

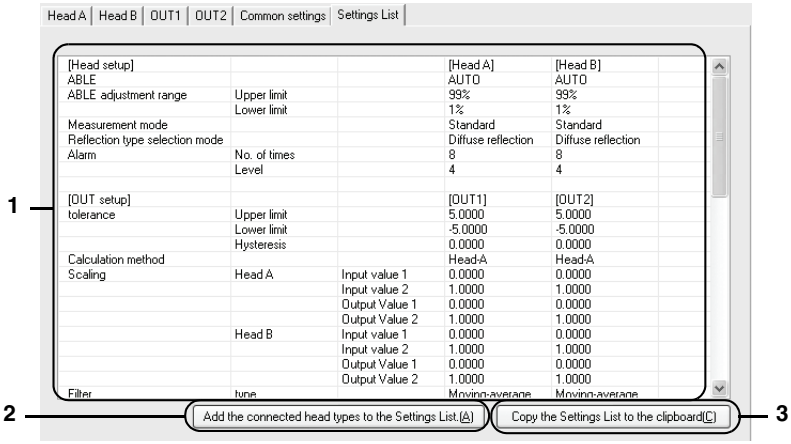


3 Click the [Paste common settings] button.

Settings List

You can confirm the items that have been set, within the program.

Click the [Settings List] tab of the Setting window.



1. Settings List

The Settings List of the selected program is displayed.

2. Adding the connected head to types the Settings List

Model of head that is connected to the controller can be added to the Settings List by using this function.

3. Copy the Settings List to the clipboard

This function is used to copy the setting to the clipboard. It is useful when pasting the settings to other applications.

Reference

- The setting that is copied to the clipboard will become the tab-delimited format when it is pasted to other applications.
- If any setting is changed after any of the following operations is made, the changed portion is displayed in blue.
  · Start of LK-Navigator
  · Sending/receiving the setting to/from Controller
  · Open/Save the Setting File

Sending/Receiving the Settings

The procedure to send or to receive the setting to or from the controller is described below.

■ Receiving the Settings

Click the [Receive settings from controller] button of the tool bar.

Reference

The same operation is obtained by selecting [Receive Settings from Controller] from [Communication] menu bar.

■ Sending the Settings

Click the [Send settings to controller] button of the tool bar.

Reference

The same operation is obtained by selecting [Send settings to controller] from [Communication] menu bar.

► Note

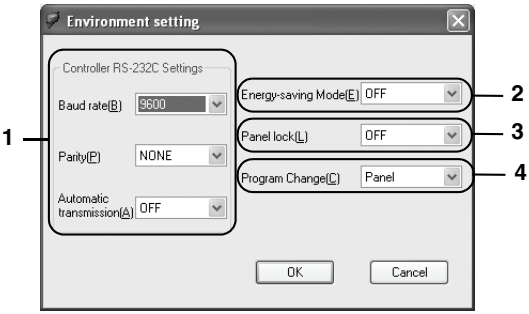
- The communication is disabled when the controller is in the "Setting mode", "Program Change mode", and "Tolerance Setting mode".
- When the controller is connected to a PC with RS-232C, and when the RS-232C setting is changed by using [Controller Environment Setting], [PC Communication Setting] of "LK-Navigator" is also changed automatically when the setting is sent to the controller.
- While executing sending/receiving setting, the controller enters the "Communication mode".

Environment Setting

The method to set the operating environment of the controller is described below.

From the main screen, click the [Controller Environment Setting] button.

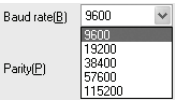
The [Environment Setting] window appears.



1. Controller RS-232C Settings

Baud rate

Set the communication speed of the controller.

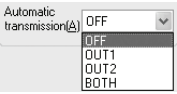


Parity

Set the type of parity.

Automatic Transmission

Set OUT that executes the automatic transmission to the RS-232C interface.



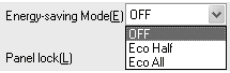
► Note

When the controller is connected to a PC with RS-232C, and when the RS-232C setting is changed by using [Controller Environment Setting], [PC Communication Setting] of "LK-Navigator" is also changed automatically after the setting is sent to the controller.

3

2. Energy-saving Mode

Set the energy-saving mode of the controller.

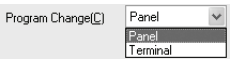


3. Panel lock

Set whether to lock the display panel that is connected to the controller.

4. Program Change

Set the method of switching the programs.



Reference

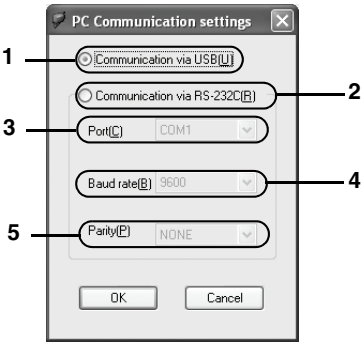
When [Terminal] is selected by [Program Change] and the setting is completed by sending/receiving the setting, the operation of “LK-Navigator” follows the Program No. that is specified by the LK-G terminal wiring.

Communication Settings

The method to set the communication between the controller and a PC is described below.

Select [PC Communication Setting] from [Communication Settings] of the menu bar.

The [PC Communication settings] window appears.



1. Communication via USB

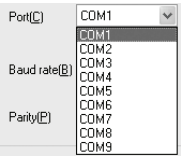
This function is used to set the communication when the USB interface is used to connect the controller to a PC.

2. Communication via RS-232C

This function is used to set the communication when the RS-232C interface is used to connect the controller to a PC.

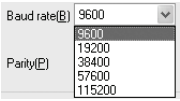
3. Port

This function is used to set the port that connects the controller to a PC when [Communication via RS-232C] is selected.



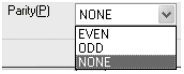
4. Baud rate

This function is used to set the baud rate in accordance with the communication speed of the controller when [Communication via RS-232C] is selected.



5. Parity

This function is used to set the method of parity check when [Communication via RS-232C] is selected.



Setting the Programs

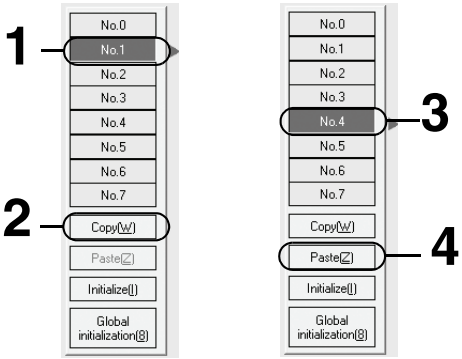
The method to copy and to initialize the program that is registered in the controller is described below.

Copy Program

Settings can be copied to the other program No. so that the nuissance to create the settings repeatedly can be avoided.

Reference

Settings of the program where settings are being copied to will be overwritten.



1 **Select the desired copy source program No. from the main screen.**

Program No. 1 is selected in this example.

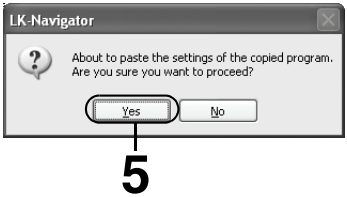
2 **Click the [Copy] button.**

3 **Confirm that the [Paste] button becomes active. Then select the program No. that is selected as the copy target.**

Program No. 4 is selected in this example.

4 **Click the [Paste] button.**

The window prompting your confirmation appears.

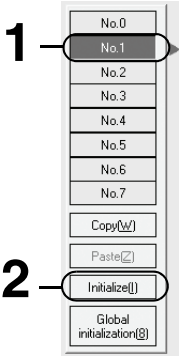


5 **Click the [Yes] button.**

The setting is copied.

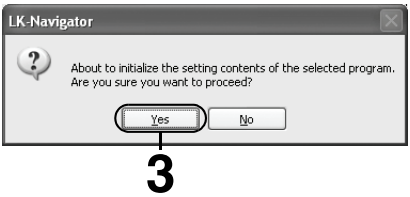
Initializing Program

Programs can be returned to the default settings when shipped from the factory by clicking [Initialize]. When a setting is initialized, the previous setting will be deleted.



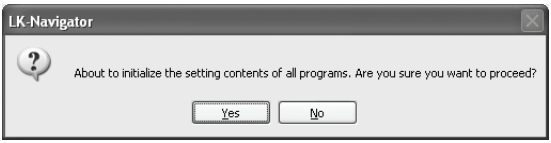
1 **Select a program No. that you want to initialize from the main screen.**
Program No. 1 is selected in this example.

2 **Click the [Initialize] button.**
The window prompting your confirmation appears.



Reference

When you want to initialize settings of all program Nos., click the [Global initialization] button. When you click the [Global initialization] button, the following window prompting your confirmation appears.



3 **Click the [Yes] button.**
The settings are initialized.

View Measurement Value

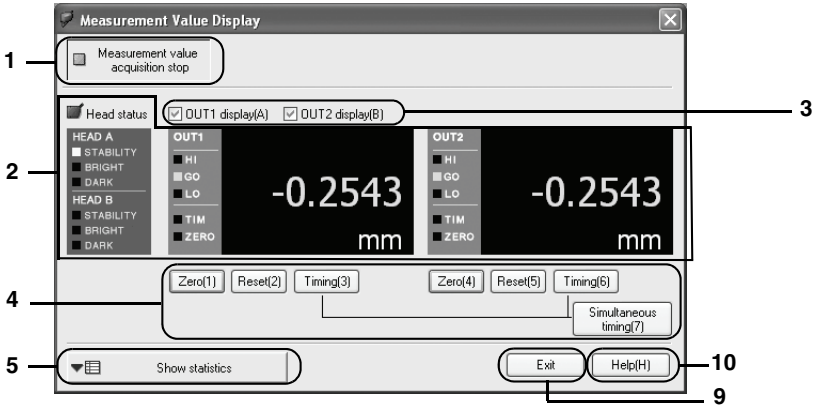
The [View Measurement Value] function that allows you to view the measured values on a PC is described below.

Click the [View Measurement Value] button of the tool bar.

Reference

The same operation is executed by selecting [View Measurement Value] from [View] on the tool bar.

The [View Measurement Value] window appears.



1. Measurement value acquisition start

Start acquisition of the measurement values from the controller.

When you click the [Measurement value acquisition start] button, it changes to the [Measurement value acquisition stop] button.

Reference

Until you click the [Measurement value acquisition start] button, any of the [Zero]/[Reset]/[Timing]/[Simultaneous Timing]/[Update Statistical Values]/[Clear Statistical Values] buttons cannot be clicked.

2. Status display

Displays the measured value/comparator output/head status.

For viewing the display, refer to "LK-G Series User's Manual" refer to "Chapter 2 Operations and Functions during Measurement" (2-1 page).

3. Display OUT1/Display OUT2

The following three types of display are available.

- OUT1 independent display : Turn ON the [Display OUT1] check box.
- OUT2 independent display : Turn ON the [Display OUT2] check box.
- OUT1 and OUT2 simultaneous display : Turn ON both of the [Display OUT1] and [Display OUT2] check boxes.

Reference

Both of the [Display OUT1] and [Display OUT2] check boxes cannot be turned OFF at the same time.

4. Displaying the measured values after auto zero, reset, and timing processing are applied.

Zero

The measured values are displayed after the present measured value is set to the reference value (zero).

For the auto zero function, refer to "LK-G Series User's Manual" refer to "Chapter 2 Operations and Functions during Measurement" (2-1 page).

Reset

The measured values displayed after the processing in a Measurement Mode are cleared.

For the Measurement Value Reset function, refer to "LK-G Series User's Manual" refer to "Chapter 3 Function Settings" (3-1 page).

Timing

Timing ON and OFF are toggled.

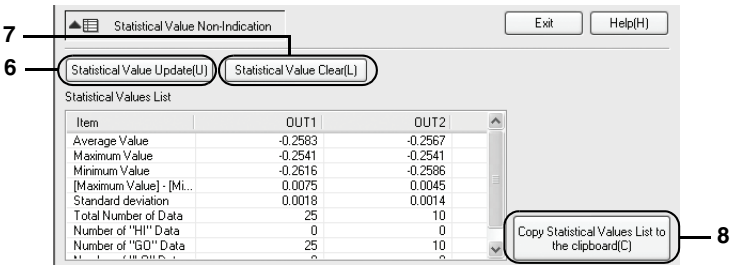
For the timing function, "LK-G Series User's Manual" refer to "LK-G Series User's Manual" refer to "Chapter 3 Function Settings" (3-1 page).

Simultaneous Timing

When OUT1 and OUT2 are displayed at the same time, the timing inputs to OUT1 and OUT2 should also be input at the same time.

5. Statistical Value Indication

When you click the [Statistical Value Indication] button, [Statistical Values List] is displayed.



6. Statistical Value Update

The statistical values are updated to the newest values.

7. Statistical Value Clear

The present statistical values are cleared.

Reference

When you click the [Statistical Value Clear] button, the statistical values of the controller are also deleted.

8. Copying the Statistical Values List to the Clipboard

This function is used to copy the Statistical Values List on display to the clipboard. It is useful when pasting the statistical value to other applications.

Reference

The statistical value that is copied to the clipboard will become the tab-delimited format when it is pasted to other applications.

9. Exit

Closes the [View Measurement Value] window.

10. Help

Opens the Help file of "LK-Navigator".

View Received Light Waveform

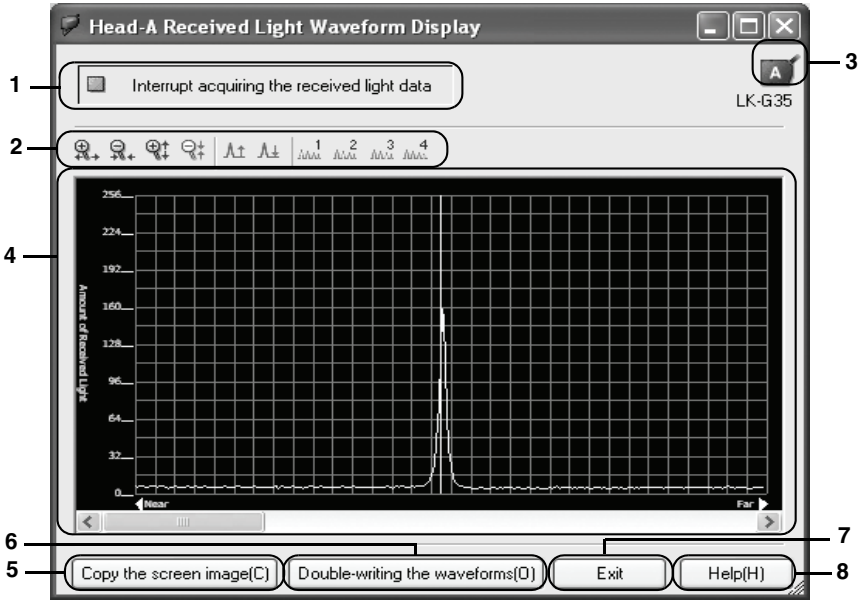
The [View Received Light Waveform] function allows you to view the measured light intensity on a PC as described below.

Click the [View Received Light Waveform] button of the tool bar. The [Head-A Received Light Waveform Display] button is clicked in this example.

Reference

The same operation as above is executed by selecting [Head-A Received Light Waveform Display] from [View] of tool bar.

The [Head-A Received Light Waveform Display] window appears.



Reference

Size of the [Head-A Received Light Waveform Display] window can be changed.
Size of the [Head-A Received Light Waveform Display] window cannot be reduced smaller than the original size of the window.

1. Start acquiring the received light data

Starts acquiring the received light data from the controller and starts displaying the newest waveform.
When you click the [Start acquiring the received light data] button, it changes to [Interrupt acquiring the received light data].

2. Waveform display change bar

Enlarge/Reduce

Waveform can be displayed after it is enlarged/reduced of its size in the vertical and horizontal directions.

Moving the display position

Waveform display position can be moved up/down.

ABLE 1 2 3 4

When [Transparent object-2] is selected by [Measurement Mode], the ABLE processing is applied to the waveforms 1 through 4, and the resultant waveforms after the processing are displayed.

For ABLE, refer to "LK-G Series User's Manual" refer to "LK-G Series User's Manual" refer to "Chapter 3 Function Settings" (3-1 page)..

3. Model display

Models of the heads that are showing their waveforms are indicated.

4. View Received Light Waveform

Displays waveform of the received light data. The white vertical line on the waveform indicates the present measuring position. The method to read the waveform is explained below.

- Horizontal axis : It indicates position of the received light waveform. As it is positioned closer to the left end, it indicates that the received light is positioned closer to the head.
- Vertical axis : It indicates intensity of the received light. As it goes higher, it indicates that the received light is more intense.
- Waveform display : Waveform after the controller applies the ABLE processing is displayed.

5. Copying the screen image

The present waveform is copied to the clipboard.

6. Double-writing the waveforms

While the waveform that is obtained at the time of clicking is kept displayed, double-writing of the present waveform over the above-mentioned waveform is executed so that the double-writing waveforms are displayed.

7. Exit

Closes the [Head-A Received Light Waveform Display] window.

8. Help

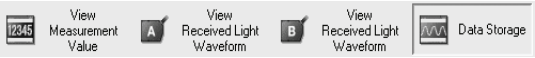
Opens the Help file of "LK-Navigator".

Data Storage Setting

The [Data Storage] function that can display the measurement result or change of data storage setting, is described below.

■ Identifying names and functions of the data storage setting window.

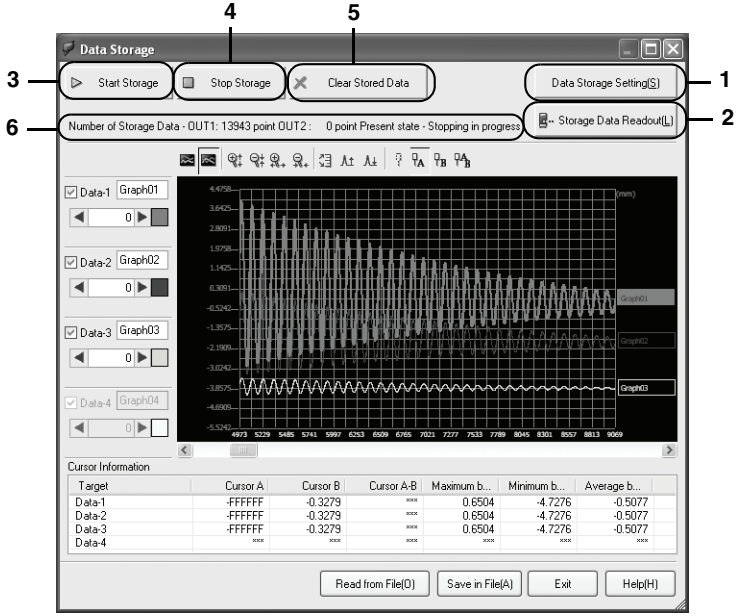
Click the [Data Storage] button of the tool bar.



Reference

The same operation as above is executed by selecting [Data Storage] from [View] on the tool bar.

The [Data Storage] window appears.



A maximum of the following four data can be displayed as the waveforms on the [Data Storage] window. The data listed below can be displayed.

- The storage data that is read out from the main unit.
- The storage data that has been stored in a PC.

Reference

Size of the [Data Storage] window can be changed.

Size of the [Data Storage] window cannot be reduced smaller than the original size of the window.

1. Data Storage Setting

Set the method to store the measured data.

Refer to "Data Storage Setting" (page 3-34).

2. Storage Data Readout

Reads out the storage data from the controller.

Refer to "Read Storage Data" (page 3-41).

3. Start Storage

The controller starts storing the data.

4. Stop Storage

The controller stops storing the data.

5. Clear Stored Data

The data that is stored in the controller is cleared.

▶ Note

Clear Stored Data clears all data that is stored in the controller.

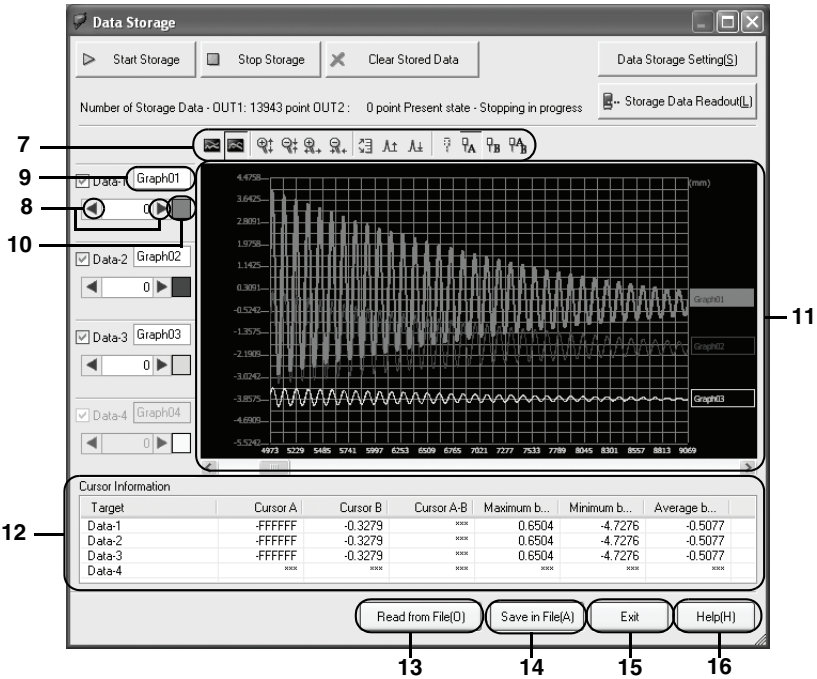
6. Status Display

The data storage status of the controller is displayed.

Reference

The statuses listed below can be displayed.

Status	Status on display	Number of storage data display
Not connected	Not connected	*****
NORMAL mode: Storage in progress	Storage in progress	Number of data that are being stored at present
NORMAL mode: Stopping in progress	Stopping in progress	Number of data that are being stored at present
When communication error occurs	*****	*****
When another system error occurs	System error	*****



7. Waveform display change bar

Display data side by side/Display waveforms by overlapping

Set whether the waveforms of the measurement values are displayed side by side or by overlapping

Enlarge/Reduce

Waveform can be displayed after it is enlarged/reduced of its size in the vertical and horizontal directions.

Automatic reduction

Reduction scale in the vertical direction is adjusted automatically and then displayed.

Moving the display position

Waveform display position can be moved up/down.

Non-display of the cursor

Set the cursor to non-display.

Selection of the cursor

Select the cursor to display.

8. Display Position Change

Move the display position of waveform.

► Note

The waveform display position can be moved only when waveform is being displayed.

9. Waveform Name

Give an arbitrary name to the waveform.

► Note

- A maximum of 16 alphanumeric characters (excluding "," (comma) and "." (period)) can be used in a name of waveform.
- Name can be given to waveform only when waveform is being displayed.

10. Change the Waveform Color

Waveform color can be set.

► Note

Waveform color can be set only when waveform is being displayed.

11. Display of Measured Value Waveform

Waveform of the readout measurement value is displayed.

12. Cursor Information

List of cursor information is displayed.

13. Read from File

Reads out the storage data from file.

14. Save in File

The stored data is saved in file.

Reference

The stored data is saved in file in the CSV format.

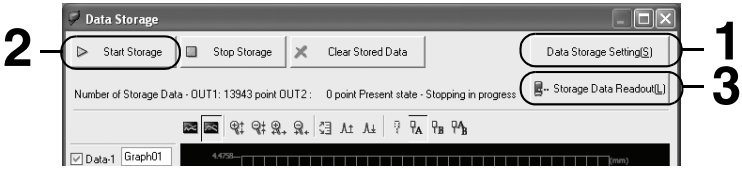
15. Exit

Closes the [Data Storage] window.

16. Help

Opens the Help file of "LK-Navigator".

■ Basic data storage procedure



1 **Click [Data Storage Setting].**
The Target OUT, the Number of storages and the Storage cycle should be set beforehand.

For the procedure of [Data Storage Setting], refer to "Data Storage Setting" (page 3-39).

2 **Click the [Start Storage] button.**
The controller starts storing the measurement data.

When the number of data reaches the set number that is specified by [Number of storages], the storage stops automatically.

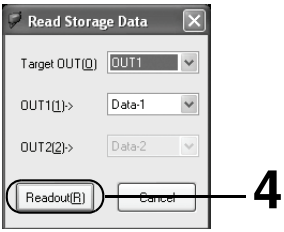
Reference

- If you want to stop the data storage in the middle of storage, click the [Stop Storage] button.
- When you click the [Start Storage] button again after the storage is stopped once, the storage is resumed from the point where the storage was stopped.

3 **Click the [Read Storage Data] button.**

For the method to set [Read Storage Data], refer to "Read Storage Data" (page 3-41).

The [Read Storage Data] window appears.

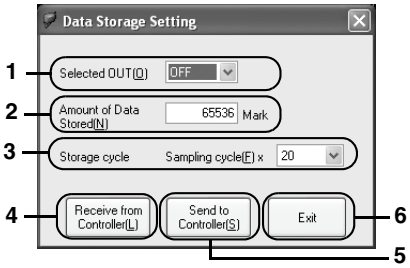


4 **Click the [Readout] button.**
The data that are accumulated in the controller are transferred to the PC.
When the bar graph indicating the transfer status reaches the right-most end, reception of the data is complete.

■ Data Storage Setting

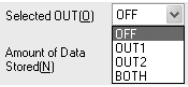
Click the [Data Storage Setting] button.

The [Data Storage Setting] window appears.



1. Selected OUT

Set the OUT that becomes the target of data storage.

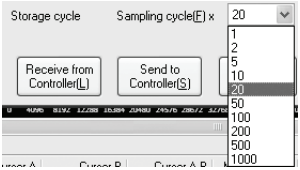


2. Amount of Data Stored

Set the amount of data to be stored.

3. Storage cycle

Set the interval at which the data is stored.



4. Receive from Controller

Receive the setting from the controller.

5. Send to Controller

Send the settings to the controller.

► Note

While executing sending or receiving the data, the controller enters the "Communication mode".

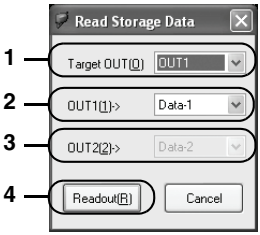
6. Exit

Closes the [Data Storage Setting] window.

■ Read Storage Data

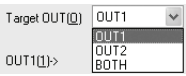
Click the [Storage Data Readout] button.

The [Read Storage Data] window appears.



1. Target OUT

Set the OUT that becomes the target of data readout.



2. OUT1

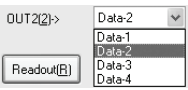
When either [OUT1] or [OUT1/OUT2] is selected by [Target OUT], the data to read out from OUT1 is selected as described below.

A waveform is displayed on the [Data Storage] window by specifying the data number that corresponds to the already set data number.

3. OUT2

When either [OUT2] or [OUT1/OUT2] is selected by [Target OUT], the data to read out from OUT2 is selected as described below.

A waveform is displayed on the [Data Storage] window by specifying the data number that corresponds to the already set data number.



4. Readout

Readout of the storage data starts.

► Note

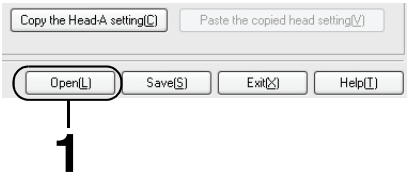
While [Readout] is being executed, the controller enters "Communication mode".

Reading and Saving the Setting File

The method to read the setting and to save it in a PC is described below.

Reading the Setting File

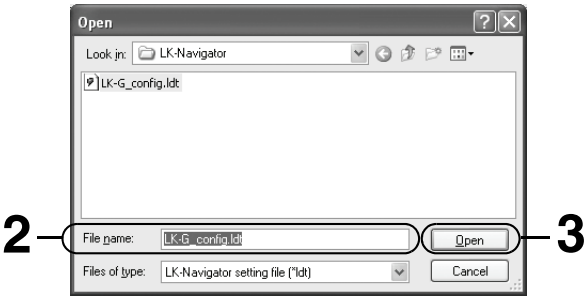
1 **Click the [Open] button of the main screen.**



Reference

The same operation is executed by selecting [Open Setting File] from [File] of the menu bar.

The [Open] window appears.



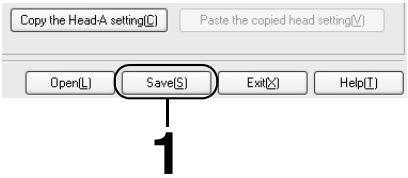
2 **Select the setting file you want to read.**

LK-G_Config.ldt is selected in this example.

3 **Click the [Open] button.**

Saving the Setting File

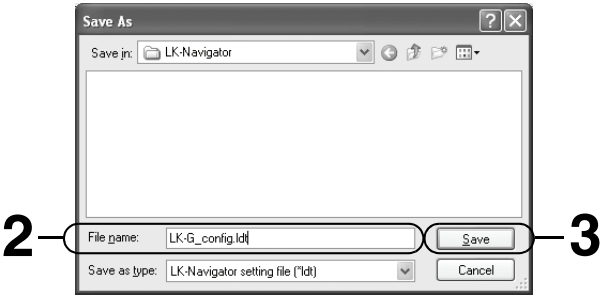
1 **Click the [Save] button of the main screen.**



Reference

The same operation is executed by selecting [Save Setting File] from [File] on the menu bar.

The [Save As] window is displayed.



2 **Open the folder where you want to save the setting file, and enter the filename to be used for saving, in [File name].**

[LK-G_config.ldt] is entered in this example.

Reference

The extension ".ldt" is automatically added to the saved file to indicate that the file is the setting file of the controller.

3 **Click the [Save] button.**